Chapter 9 • Similar Figures

Solved Review Exercise 9

MCQs

1. If two polygons are similar, then:
a) corresponding angles are equal
b) their areas are equal
c) their volumes are equal
d) corresponding sides are equal
Ans: a
Equal corresponding angles and proportional sides define similarity.

2. The ratio of areas of similar polygons is:
a) the perimeter ratio
b) the square of the corresponding-side ratio
c) the cube of the side ratio
d) the sum of corresponding sides
Ans: b
Area scales as k^2.

3. Volumes 125 cm³ and 27 cm³ give corresponding heights in ratio:
a) 3:5
b) 5:3
c) 25:9
d) 9:25
Ans: b
Height ratio $=\sqrt[3]{125}:\sqrt[3]{27}=5:3$.

4. The exterior angle of a regular pentagon is:
a) 40°
b) 45°
c) 60°
d) 72°
Ans: d
$\frac{360^\circ}{5}=72^\circ$.

5. Similar parallelograms have areas 64 and 144 cm²; a smaller side is 8 cm. Find the larger corresponding side.
a) 10 cm
b) 12 cm
c) 18 cm
d) 16 cm
Ans: b
Length ratio $=\sqrt{(\frac{144}{64})}=\frac{3}{2}$; larger side=12 cm.

6. The number of diagonals in a 9-sided polygon is:
a) 18
b) 21
c) 25
d) 27
Ans: d
$\frac{9(9-3)}{2}=27$.

7. Sphere radii are 4:5; larger surface area=500π cm². Smaller area is:
a) 256π cm²
b) 320π cm²
c) 400π cm²
d) 405π cm²
Ans: b
$500\pi(\frac{4}{5})^2=320\pi$ cm².

8. A regular polygon has exterior angle 30°. How many diagonals?
a) 54
b) 90
c) 72
d) 108
Ans: a

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$n=\frac{360}{30}=12$; diagonals$=12\times\frac{9}{2}=54$.

9. In a regular hexagon, the ratio of a diagonal to its side is:
a) √3:1
b) 2:1
c) 3:2
d) 2:3
Ans: b
For the longest diagonal, 2s:s=2:1. The wording is ambiguous: shorter diagonals give option (a), √3:1.

10. A regular polygon has interior angle 165°. How many sides?
a) 15
b) 16
c) 20
d) 24
Ans: d
Exterior=15°, so $n=\frac{360}{15}=24$.

2. The interior-angle sum of a polygon is 1080°. Find its sides.
$(n-2)180°=1080°$
$\Rightarrow n-2=6$
$\Rightarrow n=8$.

3. One similar bottle is twice as high as another. Find surface-area and capacity ratios.
Large:small surface areas = 2^2:1
= 4:1; capacities
= 2^3:1
= 8:1.

4. Every dimension of a model car is $\frac{1}{10}$ of the corresponding actual-car dimension. Find the four requested ratios.
(a) Windscreen areas: model:actual = $(1/10)^2$
= 1:100.
(b) Boot capacities: model:actual = $(1/10)^3$
= 1:1000.
(c) Widths: model:actual=1:10.
(d) Number of wheels: the count does not scale, so model:actual=1:1.

5. Similar jugs have heights 8,12,16 cm; the smallest holds $\frac{1}{2}$ litre.

Capacity scale is the cube of height scale: $(\frac{12}{8})^3=\frac{27}{8}$ and $(\frac{16}{8})^3=8$.

$V_{12}=(\frac{1}{2})\frac{27}{8}$
$=\frac{27}{16}$
= 1.6875 L
$V_{16}=(\frac{1}{2})(8)$
= 4 L.
1.6875 L (about 1.69 L) and 4 L.

6. Similar glasses have heights 7.5,9,10.5 cm; the tallest holds 343 mL.

$V_{7.5}=343(\frac{7.5}{10.5})^3$
$=343(\frac{5}{7})^3$
= 125 mL.
$V_9=343(\frac{9}{10.5})^3$
$=343(\frac{6}{7})^3$
= 216 mL.
125 mL and 216 mL respectively.

7. For similar model and actual cars, model:actual door areas=1:2500, the actual width is 150 cm and the model rear-window area is 3 cm².
(a) Length ratio=√(1:2500)=1:50.
(b) Petrol-tank capacity ratio=$(1/50)^3$=1:125000.

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(c) Model width = $\frac{150}{50}$
= 3 cm.
(d) Actual rear-window area = 3×2500
= 7500 cm².

8. The areas of two similar coffee-jar labels are in ratio 144:169. Find the ratios of jar heights and capacities.

(a) Heights = √144:√169
= 12:13.
(b) Capacities = $12^3:13^3$
= 1728:2197.

9. A repeating pattern contains one regular hexagon, six squares and six equilateral triangles, all of side $\frac{1}{2}$ m. Find its area.

Hexagon area = $(\frac{3\sqrt{3}}{2})\frac{1}{2}^2$
= $\frac{3\sqrt{3}}{8}$ m².
Six squares=$6(\frac{1}{2})^2=\frac{3}{2}$ m²; six triangles=$6(\frac{\sqrt{3}}{4})\frac{1}{2}^2=\frac{3\sqrt{3}}{8}$ m².
Total=$\frac{3}{2}\frac{+3\sqrt{3}}{4}\approx 2.79904$ m².
$\frac{6+3\sqrt{3}}{4}\approx 2.80$ m² . The printed 6.69 m² key does not follow from the stated polygons and side length.

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